| | | |
|---|---|---|
| | nucleus.) | |
| **8ai** | work function = $hc/\lambda_0$ where $\lambda_0$ is the threshold wavelength of gallium arsenide<br>$1.4 \times 1.60\times10^{-19} = 6.63\times10^{-34}\ (3.00\times10^{8}) / \lambda_0$<br>$\lambda_0 = 8.879\times10^{-7}$<br>$= 890$ nm | C1<br><br>A1 |
| **8aii** | From Wien's law,<br>$\lambda_{peak} = b/T$<br>$= 2.898\times10^{-3} / (273.15 + 37)$<br>$= 9.655\times10^{-6}$ m $= 9655$ nm<br><br>Since the peak wavelength of radiation from a human body is <u>larger than</u> the threshold wavelength, (it has lower energy than work function) and **<u>no photoelectron</u>** can be emitted. | <br><br><br>M1<br><br>A1 |
| **8b** | The photoelectron was accelerated by an electric potential difference of 150 V.<br><br>Gain in kinetic energy of photoelectron = work done by electric field on photoelectron<br>$\frac{1}{2} m v^2 - 0 = e\Delta V$<br>$v = \sqrt{\frac{2e\Delta V}{m}}$<br>$= \sqrt{\frac{2(1.60\times10^{-19})150}{9.11\times10^{-31}}}$<br>$= 7.2587\times10^{6}$<br>$= 7.3\times10^{6}$ m s$^{-1}$ | <br><br><br><br><br>C1<br><br>A1 |
| **8ci** | It ensures that every incoming electron strikes the channel wall upon entry / does not pass straight through without striking the wall | B1 |
| **8cii** | No. of wall collisions, $n = \frac{L}{2d}\sin\theta$<br>$= \frac{480}{2(8.0)}\sin 13°$<br>$= 6.75 = 6$<br><br>Since 3 electrons are emitted at each collision site,<br>electron gain = $(3)^6 = 729$ | <br><br><br>M1<br><br>A1 |
| **8d** | 1. The electrons need to <u>have high energy</u> in order to penetrate deep into the phosphor lattice.<br><br>2. High p.d. across a short distance ensures the electron <u>do not spread laterally</u> and illuminate adjacent spots on the screen. | B1<br><br>B1 |

| | | |
|---|---|---|
| **8ei** | minimum energy of each high-energy electron hits the phosphor screen<br>$= e\Delta V = 4500$ eV<br>(assuming that the energy of the electron emerging from MCP is zero)<br><br>Since each high-energy electron can create 450 electron-hole pairs,<br>average energy to create one electron-hole pair = 4500 / 450<br>= 10 eV | C1<br><br><br><br><br>A1 |
| **8eii** | total useful energy emitted as light = $nhc/\lambda$<br>$= 350\ (6.63\times10^{-34})(3.00\times10^{8}) / (545\times10^{-9})$<br>$= 1.2773\times10^{-16}$<br><br>minimum energy of incident electron = 4500 eV = $4500\ (1.60\times10^{-19})$<br>$= 7.20\times10^{-16}$<br><br>maximum internal quantum efficiency = total useful energy / minimum input energy<br>$= 1.2773\times10^{-16} / 7.20\times10^{-16}$<br>$= 0.1774 = 0.17$ | <br><br><br><br><br><br><br><br>C1<br>A1 |
| **8eiii** | For two spots to be distinguishable, the angular separation has to be greater than the Rayleigh criterion.<br><br>Rayleigh criterion, $\theta \approx \lambda / b$<br>$= 545\times10^{-9} / 7.0\times10^{-3}$<br>$= 7.79\times10^{-5}$ rad<br><br>Since the angular separation of two spots is <u>larger than the Rayleigh criterion</u> of the eye, they <u>can be distinguished</u> by the observer. | <br><br><br><br><br>M1<br><br><br>A1 |

**8f**

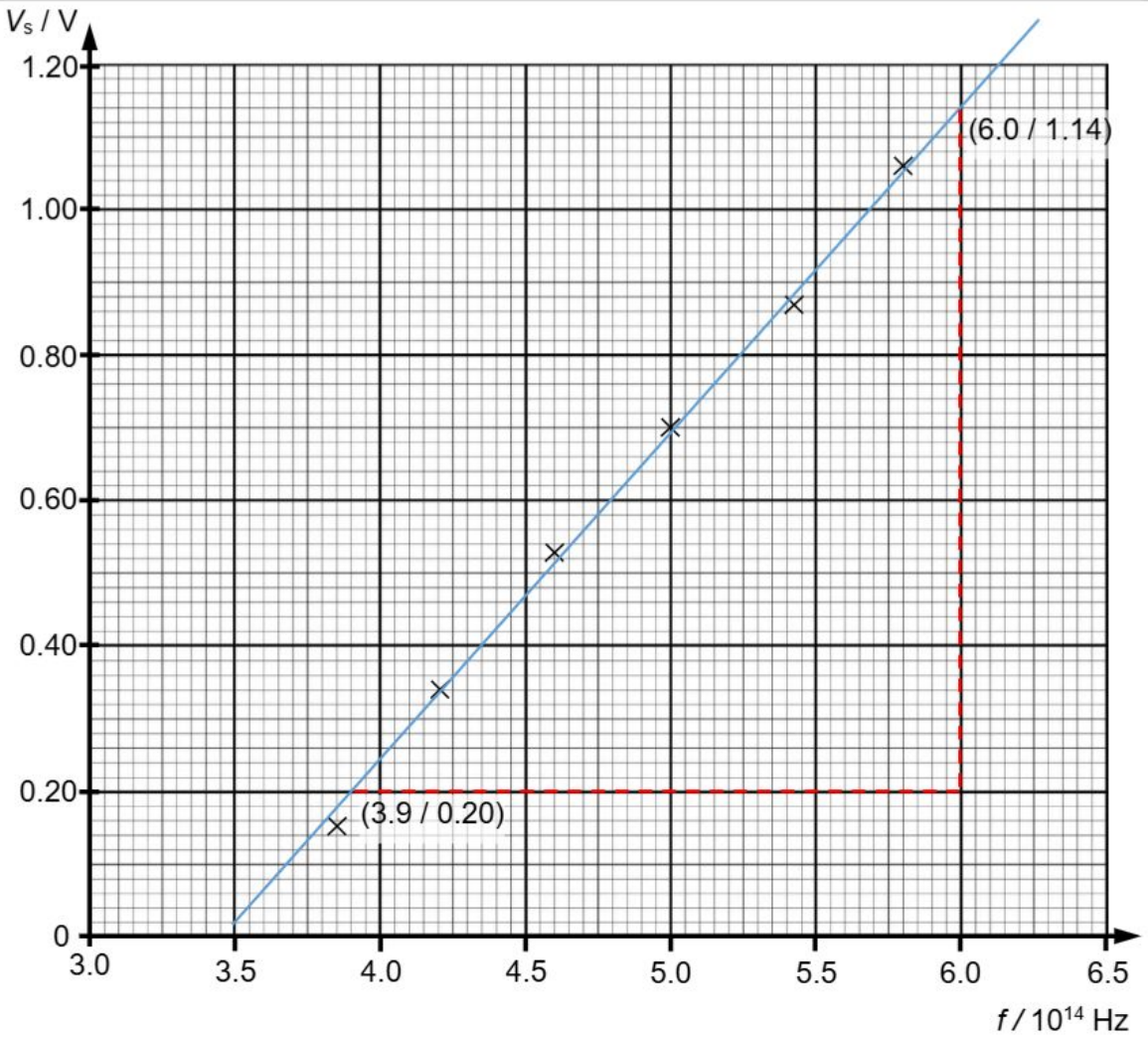



**Figure 8.3**

draw best-fit line with fair scatter of points above and below the whole line, as well as top and bottom half of the line. M1

gradient = $\frac{1.14 - 0.20}{(6.0-3.9) \times 10^{14}}$ M1

$= 4.476 \times 10^{-15}$

Rearranging the equation, we have

$$V_s = (h/e)f - \phi/e$$

gradient = $h/e$

$4.476 \times 10^{-15} = h / (1.60 \times 10^{-19})$

$h = 7.1616 \times 10^{-34}$

$= 7.2 \times 10^{-34}$ J s A1